CH 9 — Critical Points

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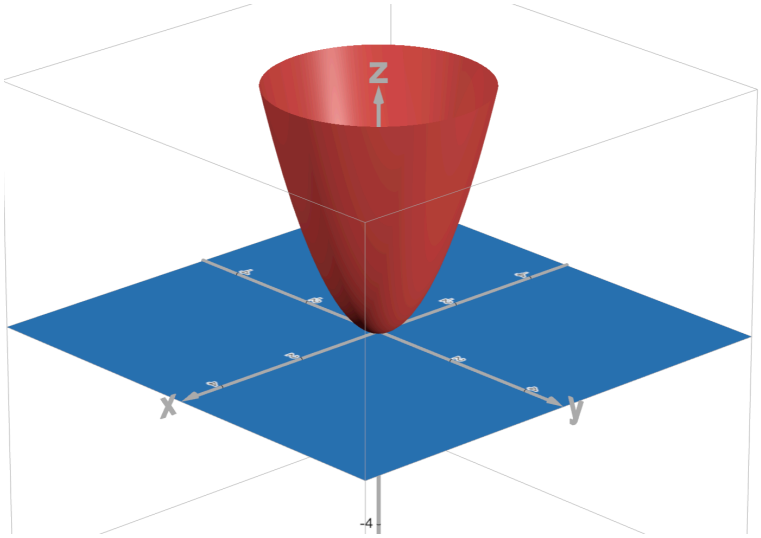
9.1 Critical Points and Local Extrema

Recall: In single -variable calculus, local extrema occur when $f'(x) = 0$ or when $f'(x)$ does not exist

> 📘 **Definition — Local Extrema**
>
> In two dimensions, (a, b) is a
>
> - **Local Maximum** of f if $f(x, y) \leq f(a, b)$ in some neighborhood of $f(a, b)$
> - **Local Minimum** of f if $f(x, y) \geq f(a, b)$ in some neighborhood of $f(a, b)$

Examples:

Local Minimum:



> 📐 **Theorem — Local Extrema**
>
> If $f(a, b)$ is a local extrema of $f(x, y)$, then
>
> $$f_x(a, b) = 0 \text{ or DNE}$$
>
> and
>
> $$f_y(a, b) = 0 \text{ or DNE}$$

Proof

Consider $g(x) = f(x, b)$. If (a, b) is a local extrema of f, $x = a$ must be a local extrema of $g(x)$. Therefore, $g'(a) = 0 \text{ or DNE} = f_x(a, b)$.

Similar argument can be applied for $g(x) = f(a, y)$

□

Definition — Critical Points

A point $(a, b) \in \text{dom} f$ is a critical point of f if

$$f_x(a, b) = 0 \text{ or DNE}$$

and

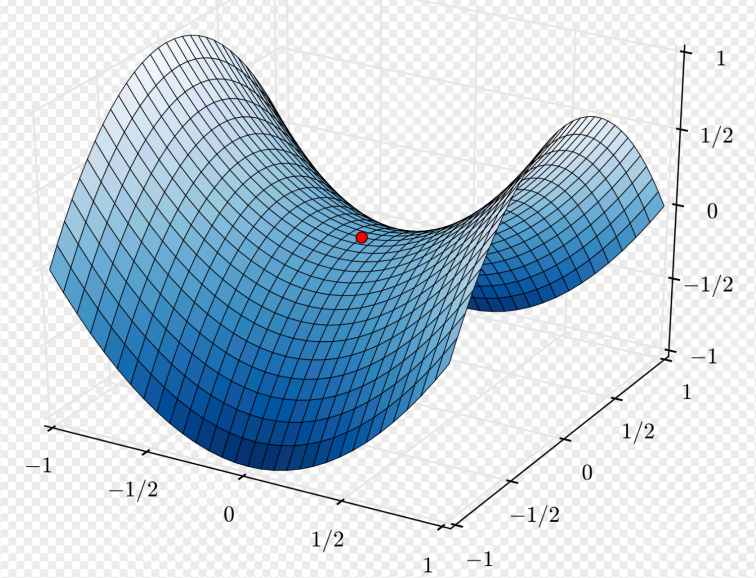
$$f_{y(a,b)} = 0 \text{ or DNE}$$

A local extrema must occur at a critical point BUT the reverse is not true.

Definition — Saddle Point

A critical point (a, b) is a **saddle point** of $f(a, b)$ if in every neighborhood of $(a, b) \exists (x_1, y_1)$ and (x_2, y_2) s.t.

$$f(x_1, y_1) > f(a, b) > f(x_2, y_2)$$



Example:

Find all critical points of $f(x, y) = x^2y - 2xy^2 + 3xy + 4$

$f_x = 2xy - 2y^2 + 3y, f_y = x^2 + 4xy + 3x$. It is two continuous functions, so the critical points must have $f_x = f_y = 0$

$2xy - 2y^2 + 3y = x^2 - 4xy + 3x = y(2x - 2y + 3) = x(x - 4y + 3)$

$y_1 = 0, x_1 = 0 : (0, 0)$

$y_2 = 0, x + 3 = 0 \implies x_2 = -3 : (-3, 0)$

$2x + 2y + 3 = 0, x_3 = 0 \implies -2y + 3 = 0, y_3 = \frac{3}{2} : \left(0, \frac{3}{2}\right)$

$2x - 2y + 3 = 0, x - 4y + 3 = 0 \implies x_4 = -1, y_4 = \frac{1}{2} : \left(-1, \frac{1}{2}\right)$

We have 4 critical points: $(0, 0), (-3, 0), \left(0, \frac{3}{2}\right), \left(-1, \frac{1}{2}\right)$

9.2 Second Derivative Test

9.3 Proof of the Second Partial Derivative Test